

المحاضرة الأولى

حسبان ؟ " التكامل "

II التكامل الغير محدد :-

$$\frac{dF}{dx} = g(x) \rightarrow \int g(x) dx = F(x) + C$$

متغير ثابت التكامل

المشتقة العكسية للدالة $\frac{dF}{dx}$

التكامل عملية عكسية للتفاضل

III القواعد الأساسية للتكاملات :-

1- $\int dF(x) \rightarrow F(x) + C$

2- $\int dx \rightarrow x + C$

3- $\int [F(x) \pm g(x)] dx \rightarrow \int F(x) dx \pm \int g(x) dx \rightarrow F(x) + \dots + C$

4- $\int \frac{F'(x)}{F(x)} dx \rightarrow \ln F(x) + C$ تكامل أي دالة كسرية، ~~تعاين المقام~~ لو غارت المقام لو البسط مشتقة المقام

5- $\int e^{F(x)} \cdot F'(x) dx \Rightarrow e^{F(x)} + C$

6- $\int F^n(x) \cdot F'(x) dx \Rightarrow \frac{F^{n+1}(x)}{n+1} + C$

7- $\int x^n dx \Rightarrow \frac{x^{n+1}}{n+1} + C$ حالة خاصة *

8- $\int \frac{1}{x} dx \Rightarrow \ln x + C$ حالة خاصة *

$$\textcircled{1} - \sin^2 x + \cos^2 x = 1$$

$$\textcircled{2} - \sin(x + \theta) = \sin x \cdot \cos \theta + \cos x \cdot \sin \theta$$

$$\textcircled{3} - \cos(x + \theta) = \cos x \cdot \cos \theta - \sin x \cdot \sin \theta$$

$$\textcircled{4} - \tan(x + \theta) = \frac{\tan x \pm \tan \theta}{1 - \tan x \cdot \tan \theta}$$

$$\textcircled{5} - \cos(2x) \Rightarrow 1 - 2\sin^2 x$$

$$\textcircled{6} - \cos(2x) \Rightarrow 2\sin^2 x - 1$$

$$\textcircled{7} - \cos(2x) \Rightarrow \cos^2 x - \sin^2 x$$

$$\textcircled{8} - \sin(2x) \Rightarrow 2\sin x \cdot \cos x$$

$$\textcircled{9} - 1 + \tan^2 x \Rightarrow \sec^2 x$$

$$\textcircled{10} - \int \sin f(x) \cdot \frac{df}{dx} dx \Rightarrow -\cos f(x) + C$$

$$\textcircled{11} - \int \cos f(x) \cdot \frac{df}{dx} dx \Rightarrow \sin f(x) + C$$

$$\textcircled{12} - \int \sec^2 f(x) \cdot \frac{df}{dx} dx \Rightarrow \tan f(x) + C$$

$$\textcircled{13} - \int \csc^2 f(x) \cdot \frac{df}{dx} dx \Rightarrow -\cot f(x) + C$$

$$\textcircled{14} - \int \sec f(x) \tan f(x) \cdot \frac{df}{dx} dx \Rightarrow \sec f(x) + C$$

$$\textcircled{15} - \int \csc f(x) \cot f(x) \cdot \frac{df}{dx} dx \Rightarrow -\csc f(x) + C$$

$$\int \frac{dx}{\sqrt{x^2 - a^2}} = \cosh^{-1} \frac{x}{a} + C$$

$$\int \frac{dx}{\sqrt{a^2 + x^2}} = \sinh^{-1} \frac{x}{a} + C$$

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \frac{1}{a} \tanh^{-1} \frac{x}{a} + C$$

[3] أوجد ناتج التكاملات التالية:

$$1 - \int \tan(x) dx = \int \frac{\sin x}{\cos x} dx = -\ln |\cos x| + C$$

$$2 - \int \cot(x) dx = \int \frac{\cos x}{\sin x} dx = \ln |\sin x| + C$$

$$3 - \int \sec(x) dx = \int \sec(x) \cdot \frac{\sec x + \tan x}{\sec x + \tan x} dx$$

$$= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx$$

$$= \ln |\sec x + \tan x| + C$$

$$4 - \int \csc x = \int \csc x \cdot \frac{\csc x + \cot x}{\csc x + \cot x} dx$$

$$= \int \frac{\csc^2 x + \csc x \cot x}{\csc x + \cot x} dx$$

$$= -\ln |\csc x + \cot x| + C$$

$$5 - \int \frac{x^2}{1+x} dx = \int \frac{x^2 - 1 + 1}{1+x}$$

$$= \int \frac{x^2 - 1}{1+x} dx + \int \frac{1}{1+x} dx$$

$$= \int \frac{(x+1)(x-1)}{\cancel{1+x}} dx + \int \frac{1}{1+x} dx$$

$$= \frac{1}{2} x^2 + x + \ln(1+x) + C$$

$$6 - \frac{dx}{1+\cos x} = \int \frac{1}{1+\cos x} \cdot \frac{1-\cos x}{1-\cos x} dx$$

$$= \int \frac{1-\cos x}{1-\cos^2 x} dx$$

$$= \int \frac{1-\cos x}{\sin^2 x} dx$$

$$= \int \frac{1}{\sin^2 x} dx - \int \frac{\cos x}{\sin^2 x} dx$$

$$= \int \csc^2 dx - \int \sin^{-2} x \cdot \cos x dx$$

$$= -\cot x + \sin^{-1} x + C$$

أوجد قيمة هذا التكامل :

$$1 - \int \frac{dx}{1+e^x} = \int \frac{1}{1+e^x} \cdot \frac{e^{-x}}{e^{-x}} dx$$

$$= - \int \frac{e^{-x}}{e^{-x}+1} dx$$

$$= - \ln(e^{-x}+1) + C$$

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$$2 - \int \sin^3 x dx = \int \sin^2 x \cdot \sin x dx$$

$$= \int (1 - \cos^2 x) \cdot \sin x dx$$

$$= \int \sin x dx - \int \cos^2 x \sin x dx$$

$$= -\cos x + \frac{1}{3} \cos^3 x + C$$

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$$\int \sin^4 x dx = \int \sin^3 x \cdot \sin x dx$$

$$\sin^2(x) + \cos^2(x) = 1$$

$$\cos(2x) = 1 - 2\sin^2(x)$$

$$= 2\cos^2(x) - 1$$

$$\sin(2x) = 2\sin(x)\cos(x)$$

$$1 + \tan^2(x) = \sec^2(x)$$

$$1 + \cot^2(x) = \csc^2(x)$$

$$\sin(x)\cos(y) = \frac{1}{2}\left[\frac{\sin x - y}{2} + \sin\left(\frac{x+y}{2}\right)\right]$$

$$\sin(x)\sin(y) = \frac{1}{2}\left[\cos\frac{x-y}{2} - \cos\frac{x+y}{2}\right]$$

$$\cos(x)\cos(y) = \frac{1}{2}\left[\cos\left(\frac{x-y}{2}\right) + \cos\left(\frac{x+y}{2}\right)\right]$$

$$\begin{aligned}\int \frac{x+1}{\sqrt{x}} dx &= \int \frac{x}{\sqrt{x}} + \frac{1}{\sqrt{x}} dx \\ &= \int x^{1/2} dx + \int x^{-1/2} dx \\ &= \frac{2}{3} x^{3/2} + 2x^{1/2} + C\end{aligned}$$

$$\begin{aligned}\int \sec(x) dx &= \sec(x) \cdot \frac{\sec(x) \cdot \tan(x)}{\sec^2(x) \cdot \tan(x)} \\ &\quad - \frac{\sec^2(x) \cdot \sec(x) \tan(x)}{\sec(x) \tan(x)} \\ &= \ln(\sec(x) \tan(x)) + C\end{aligned}$$

$$\begin{aligned}\int \frac{\sqrt{\ln(x)}}{x} dx &= \int (\ln x)^{1/2} \cdot \frac{1}{x} dx \\ &= \frac{2}{3} (\ln x)^{3/2} + C\end{aligned}$$

$$\begin{aligned}\int \frac{e^{3/x}}{x^2} dx &= \int e^{3/x} \cdot \frac{1}{x^2} \cdot dx \\ &\quad \frac{1}{3} \int -3e^{3/x} \cdot \frac{1}{x^2} \cdot dx \\ &= -\frac{1}{3} e^{3/x} + C\end{aligned}$$

$$\int \sin^7(3x) \cos(3x) dx$$

$$= \frac{1}{8} (\sin(3x))^8 + C$$

$$\int (x^2 + 1)^2 \cdot x \cdot dx$$

$$\frac{1}{2} \int 2(x^2 + 1)^2 \cdot x$$

$$= \frac{(x^2 + 1)^3}{3} + C$$